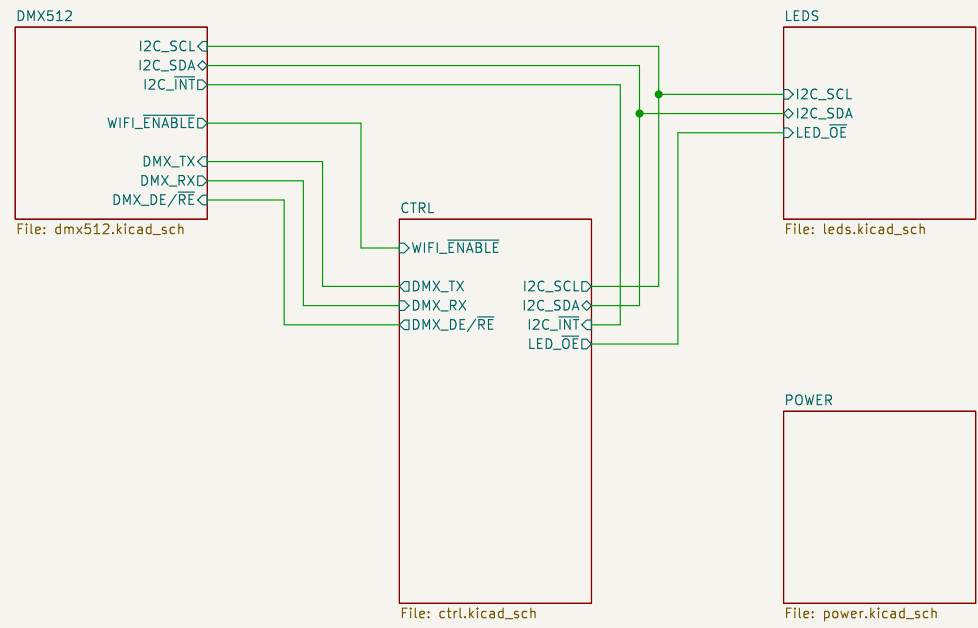


LED Beacon

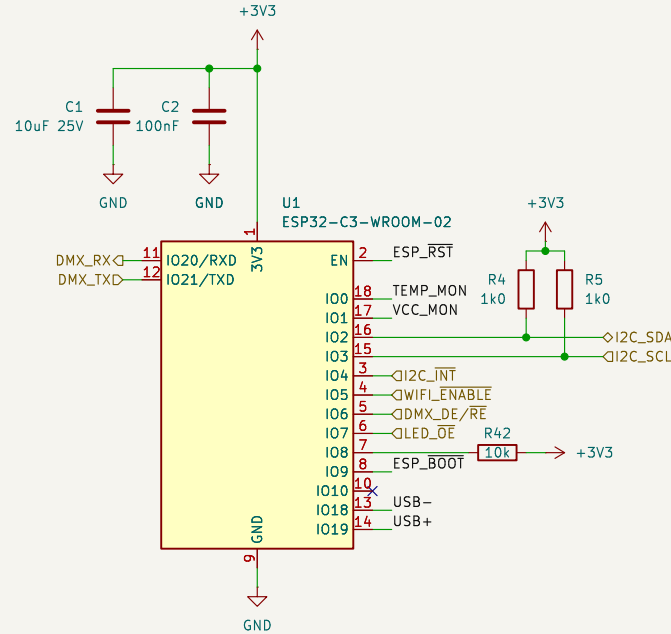
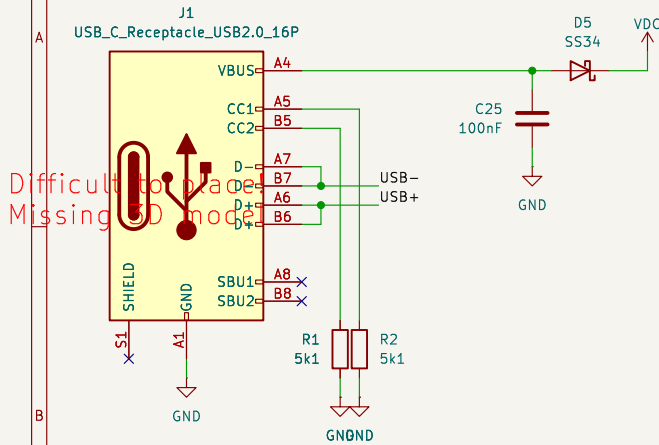


- H1 MountingHole
- H2 MountingHole
- H3 MountingHole
- H4 MountingHole

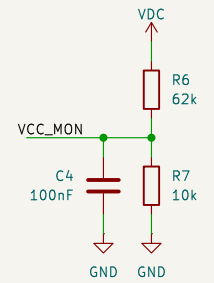
Wires & Bits		
Sheet: /		
File: emf_beacon.kicad_sch		
Title:		
Size: A4	Date: 2026-04-10	Rev: 0
KiCad E.D.A. 9.0.9		Id: 1/5

Controller

USB Programming

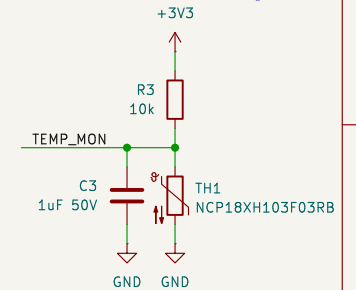


VCC Monitoring

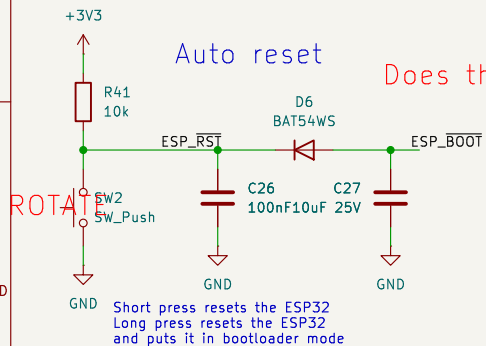


Voltage Monitor
Vmax is 17.5V
 $V_{adc} \times ((43k\Omega + 10k\Omega) + 10k\Omega) = VCC$
 $V_{adc} \times 5.3 = VCC$
 $1lsb = 4,271\mu V$

Temperature Monitoring



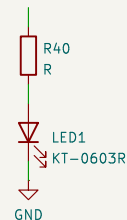
Auto reset



Does this need to be a switch?
Cheaper switch?
Side actuation?

Status LED

Status LEDs?



Wires & Bits

Sheet: /CTRL/
File: ctrl.kicad_sch

Title:

Size: A4

Date: 2026-04-10

Rev: 0

KiCad E.D.A. 9.0.9

Id: 2/5

DMX512

To XLRs

J2 Conn_01x06

J3 Conn_01x06

R8 1MΩ

C5 1nF 2kV

GND_ISO

DMX_DATA-

DMX_DATA+

DMX_AUX-

DMX_AUX+

Fail-safe Bias

ESD Protection

D1 SM712

R10 4.7k

R11 4.7k

GND_ISO

+3V3_ISO

C7 10uF 25V

C8 100nF

GND_ISO

Isolated RS485 TRX

+3V3_ISO

+3V3

U3 CA-IS3092W

C9 100nF

C10 10uF 25V

GND

GND

OUT OF STOCK!

DMX_DATA+

DMX_DATA-

DMX_TX

DMX_DE/RE

DMX_RX

R43 310k

+3V3

GND

Make a symbol for CA-IS3092W CA-IS3092VW

GPA7 & GPB2:7 for?

SW1 SW_DIP_x10

DMX_ADDR0

DMX_ADDR1

DMX_ADDR2

DMX_ADDR3

DMX_ADDR4

DMX_ADDR5

DMX_ADDR6

DMX_ADDR7

DMX_ADDR8

WIFI_EN via GPIO expander?

U2 MCP23017T-E/SS

VDD

SDA

SCL

INTB

INTA

RESET

A2

A1

A0

VSS

GND

GND

I2C Addr: 0x20

Wires & Bits

Sheet: /DMX512/

File: dmx512.kicad_sch

Title:

Size: A4

Date: 2026-04-10

Rev: 0

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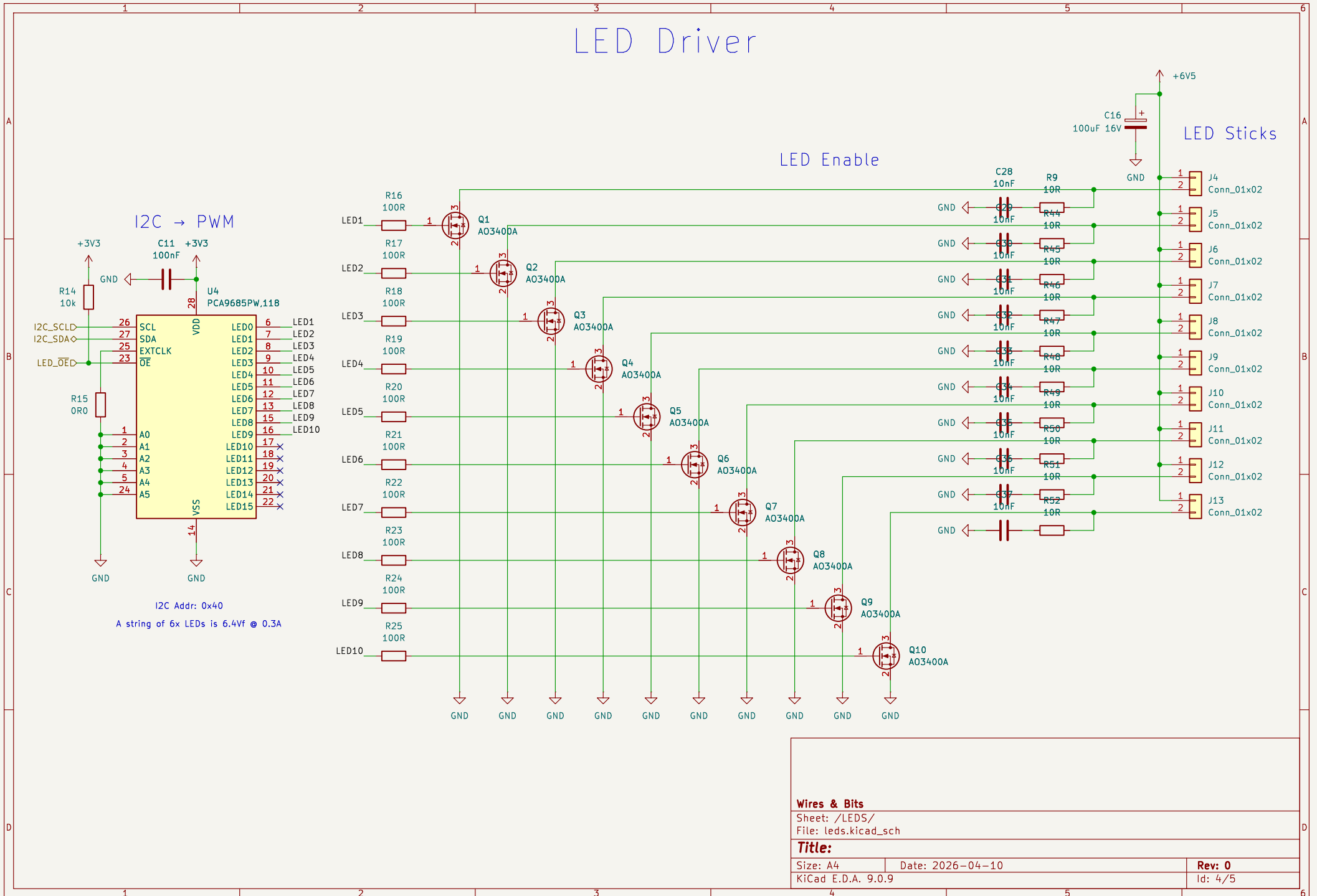
Id: 3/5

Id: 3/5

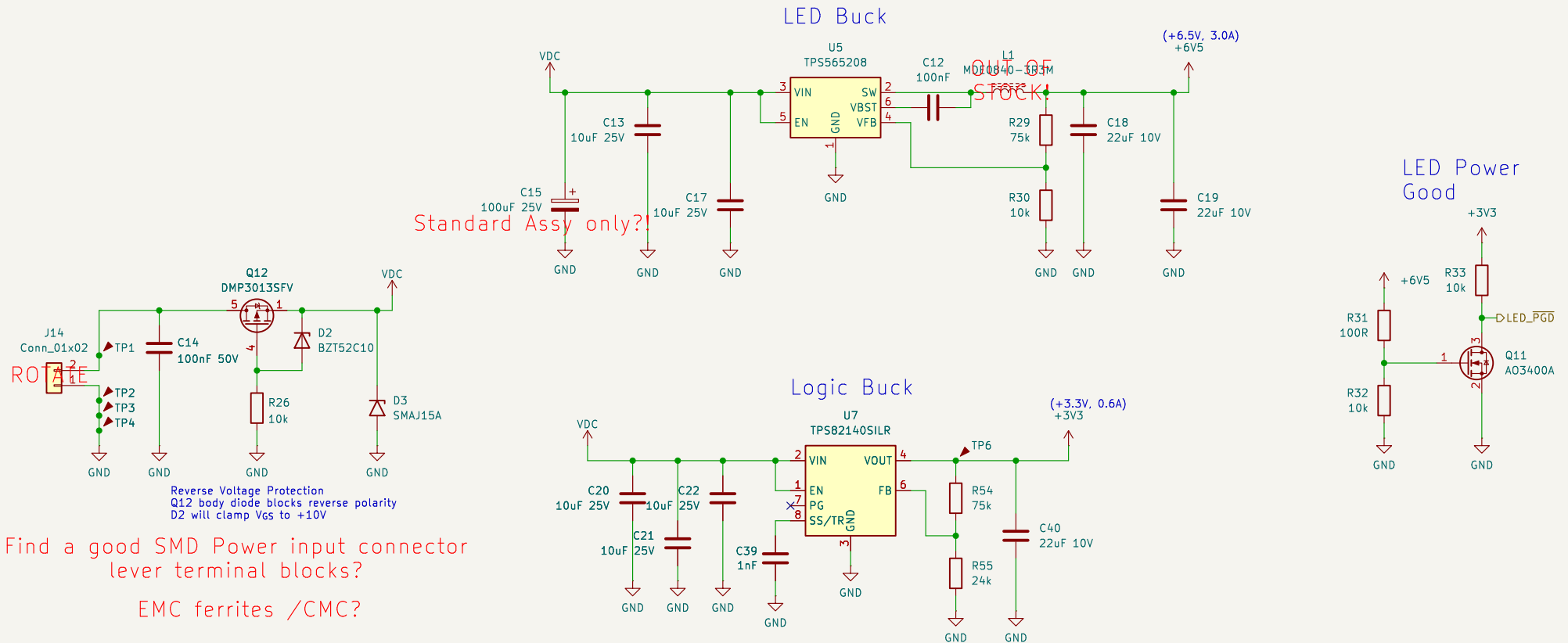
LED Driver

LED Enable

LED Sticks



Power Supplies



Wires & Bits

Sheet: /POWER/
File: power.kicad_sch

Title:

Size: A4

Date: 2026-04-10

KiCad E.D.A. 9.0.9

Rev: 0

Id: 5/5